

Henrique Lopes

Porto, Portugal

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Profile

Junior AI Engineer with hands-on experience in AI integration, Web Development, and interactive applications. Skilled in building full-stack applications with Flask, FastAPI, React, Angular, and integrating LLMs (GPT, LangChain) and ML frameworks (PyTorch, TensorFlow) into production-ready solutions. Passionate about research, prototyping, and collaborative problem-solving.

Education

Faculty of Engineering, University of Porto (FEUP)	Porto
MSc in Multimedia – Interactive Technologies and Digital Games	2024–2025
University of Minho	Braga
BSc in Engineering Physics	2017–2023

Professional Experience

INESC TEC	Porto
AI Developer Intern	2024–2025
- Developed VR applications with GPT-based conversational AI, Unity, and Python APIs, enhancing real-time user engagement. - Built RESTful and FastAPI services to integrate ML models into applications. - Implemented ML pipelines using PyTorch and TensorFlow for prototyping, experimentation, and testing. - Collaborated in multidisciplinary teams, improving system design and performance iteratively.	

Selected Projects

- ThinkAlike:** Collaborative platform with AI-assisted UI; integrated LLMs for question generation, difficulty adjustment, and text reformulation. (React, Node.js, LangChain)
- StoryShow:** Personalized recommendation platform for movies, series, and books using content-based and collaborative filtering. (PHP, MySQL, Python: scikit-learn, pandas)
- Smart Health Tracker:** Health monitoring app collecting user activity, sleep, and nutrition data; provides ML-driven personalized recommendations. (Python: Flask/FastAPI, React Native, PostgreSQL, TensorFlow)
- Virtual Moderator in VR:** Master's thesis project: developed VR environment facilitating social interactions with GPT-powered conversational AI. (Unity, Python, GPT-4, LangChain, Chroma)

Technical Skills

- Programming:** Python, JavaScript (ES6+), TypeScript, Java, C#, C, C++, Haskell, Assembly.
- Web Development:** Flask, FastAPI, PHP, React, Angular
- Databases & Data:** PostgreSQL, MySQL, MongoDB, JSON; Pandas, NumPy, scikit-learn, Chroma embeddings
- AI & ML:** PyTorch, TensorFlow, GPT, LangChain, Llama, Deepseek, RAG pipelines, ML prototyping
- DevOps & Cloud:** Docker, Docker Compose, Kubernetes, AWS CI/CD, Azure
- Tools:** Git, GitHub, VS Code, Postman, Prometheus, Grafana
- Methodologies:** Agile (Scrum, Kanban), collaborative software development

Languages

- Portuguese: Native
- English: Advanced